Systemtheorie und Regelungstechnik

TECHNISCHE UNIVERSITÄT DARMSTADT

Vorlesung 08 | Datum:



Kap. 5 (Fortsetzung): Nicht phasenminimale Systeme
Kap. 6: Stabilität

1. Nicht phasenminimale Systeme
2. Stabilität
3. Pole/Nullstellen ↔ Bode

1. Nicht phasenminimale Systeme

Totzeitglied – Beispiel

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z. B. Förderband:





1. Nicht phasenminimale Systeme

Totzeitglied: Übertragungsfunktion und Padé-Approximation

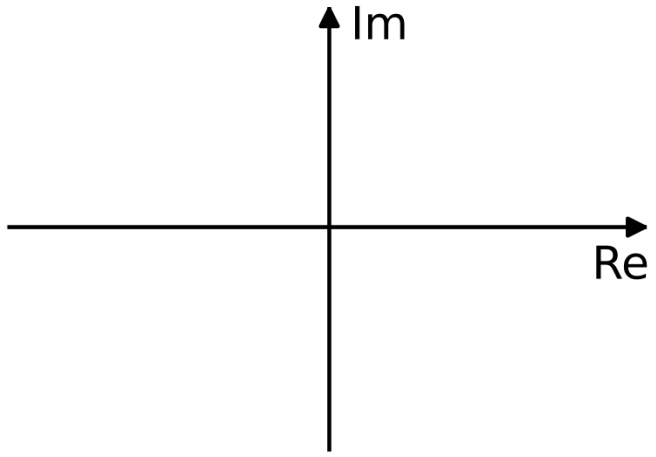
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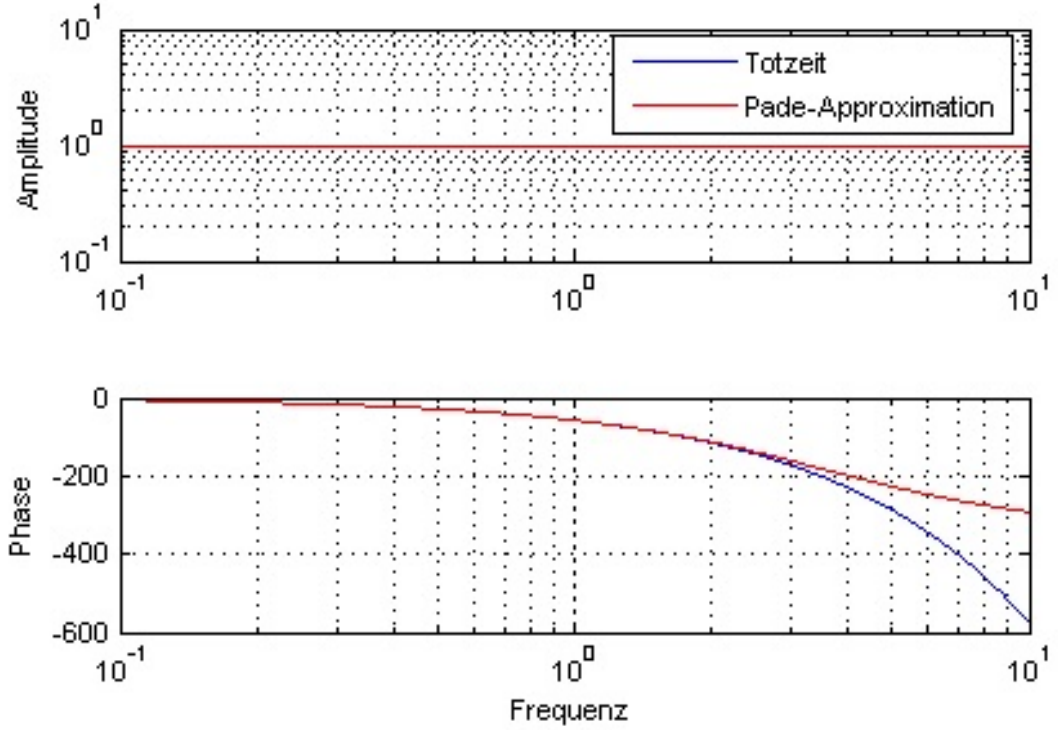
Übertragungsfunktion:

Padé-Approximation:

1. Nicht phasenminimale Systeme

Padé-Approximation: Pole/Nullstellen und Bode-Diagramm



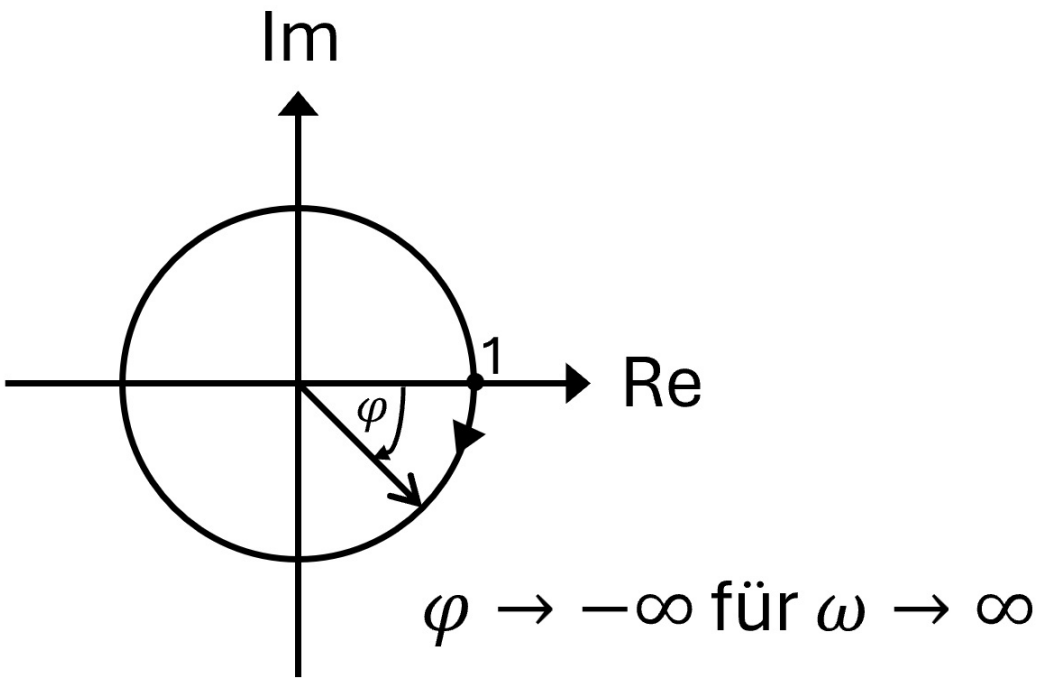


1. Nicht phasenminimale Systeme

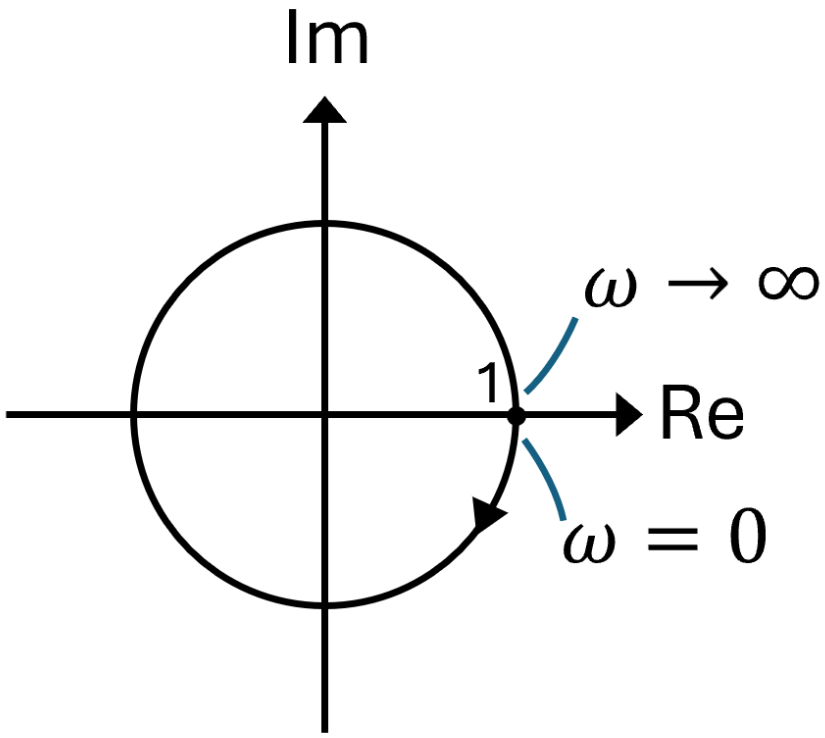
Totzeit und Padé-Approximation: Ortskurven

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Totzeit

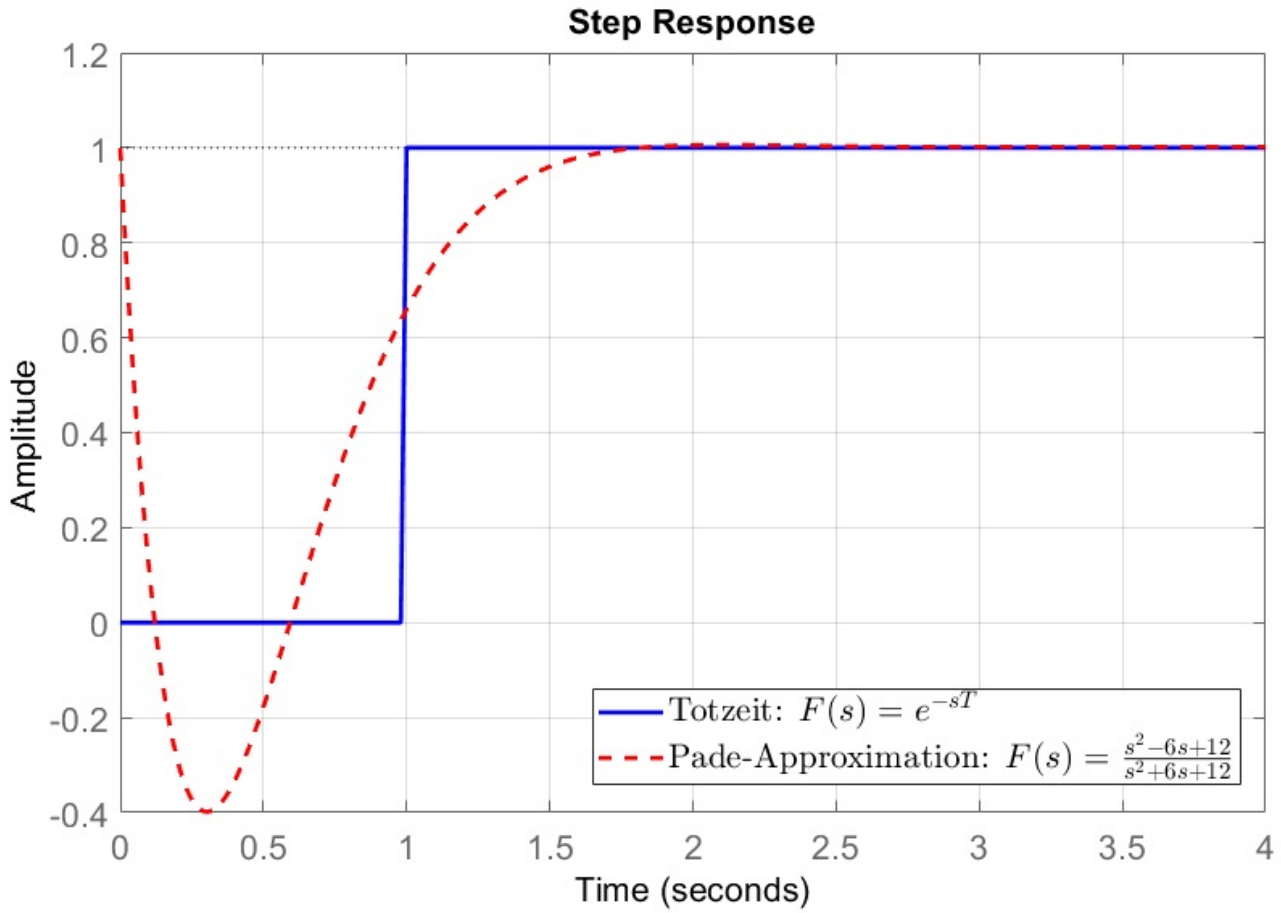


Padé 2. Ordnung



1. Nicht phasenminimale Systeme

Sprungantwort der Padé-Approximation

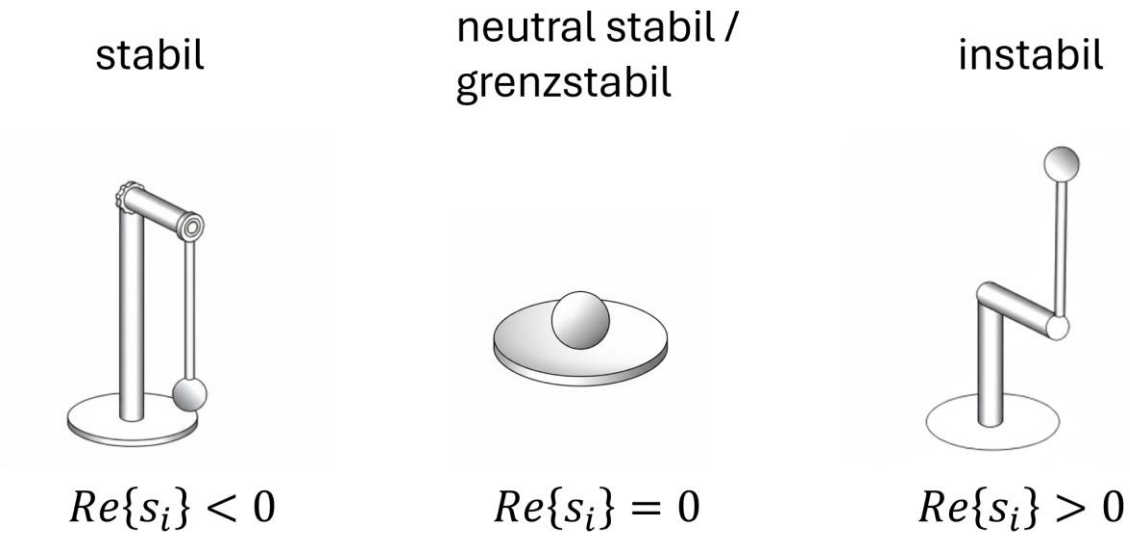


2. Stabilität

Definition

Definition:

Gewichtsfunktion $g(t)$:

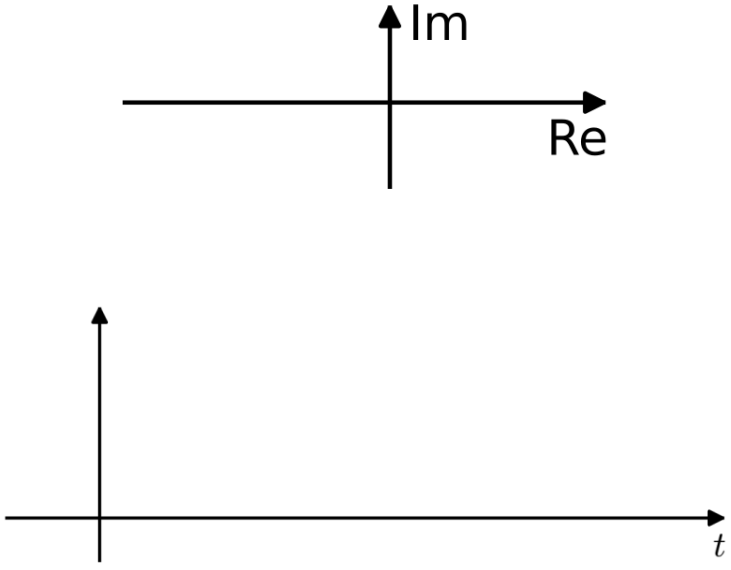


2. Stabilität

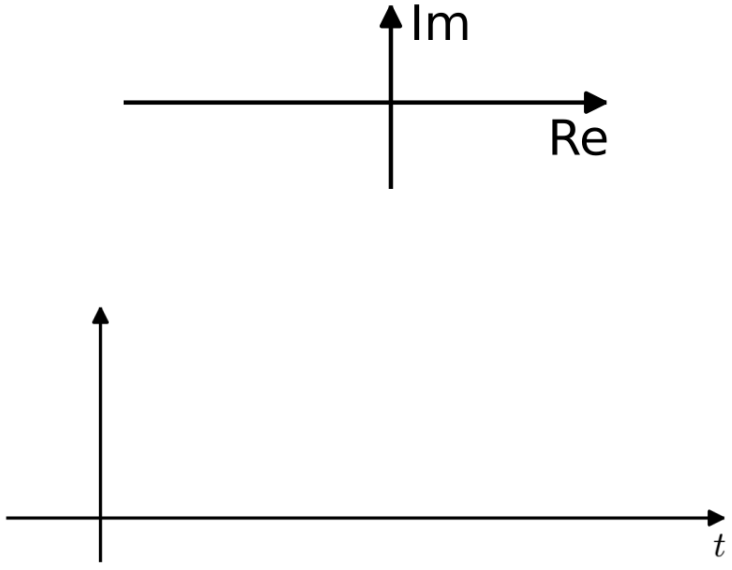
Beispiele: Gewichtsfunktion und Sprungantwort

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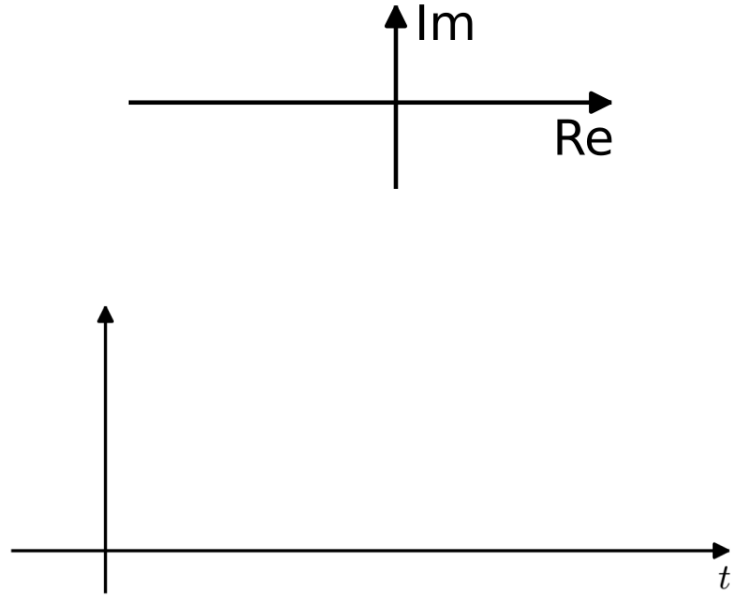
PT_1, **stabil**



I, **grenzstabil**

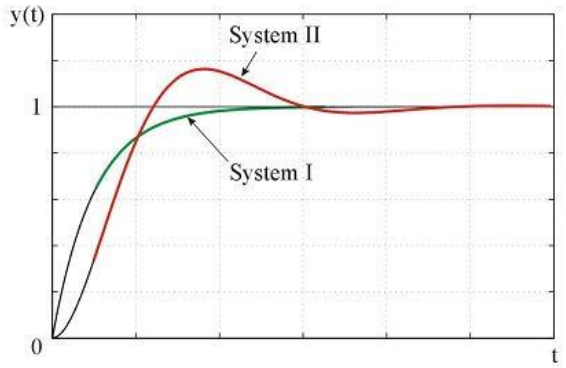


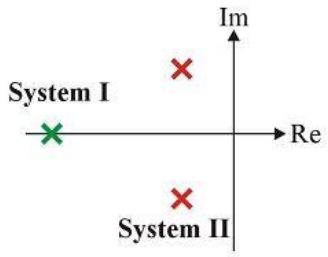
I_2, **instabil**



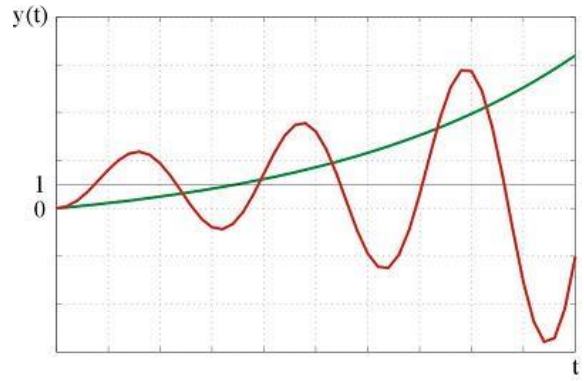
2. Stabilität

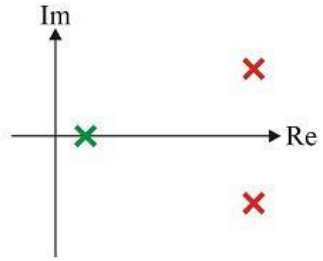
Übersicht: stabil, grenzstabil, instabil



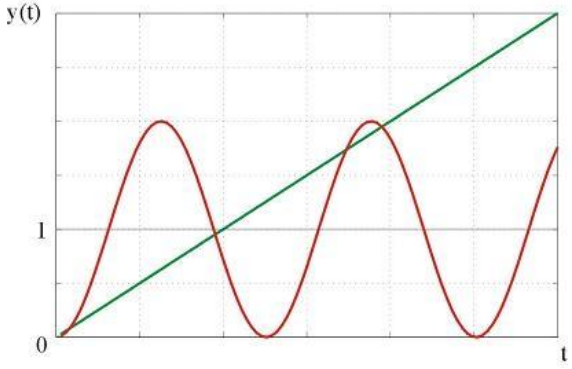


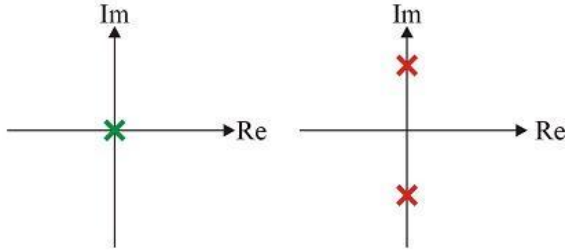
Stabil



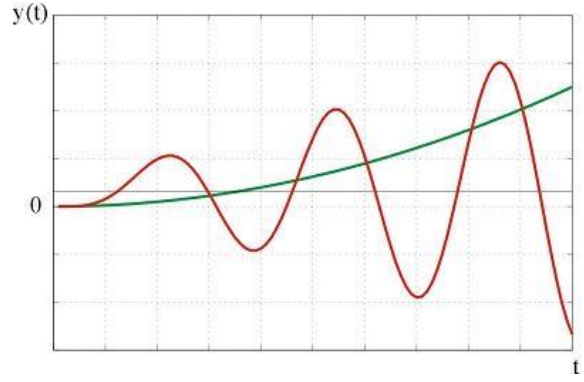


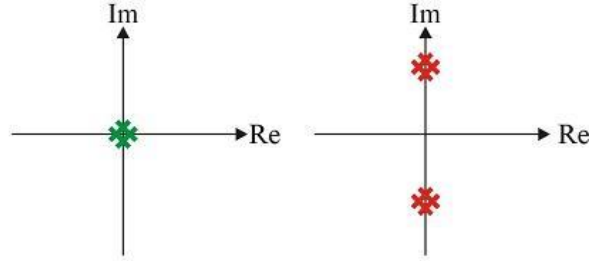
Instabil





Grenzstabil



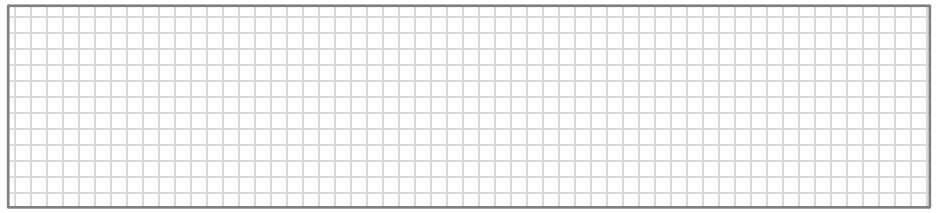


Instabil

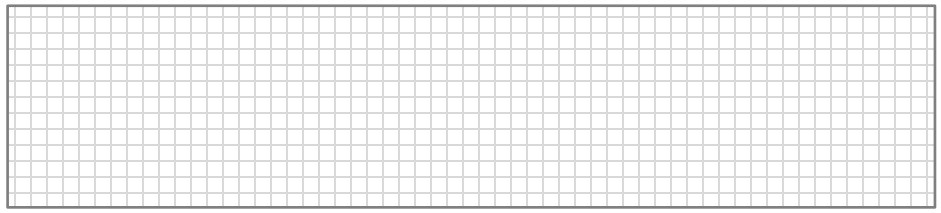
3. Pole/Nullstellen ↔ Bode

Nullstellen: Beispiel PDT_1

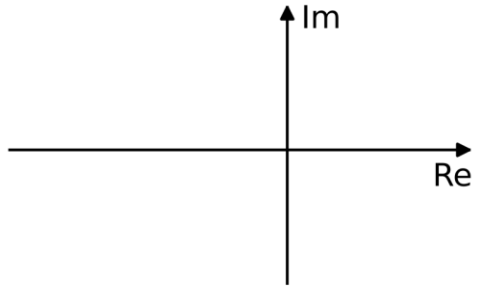
Fall a)

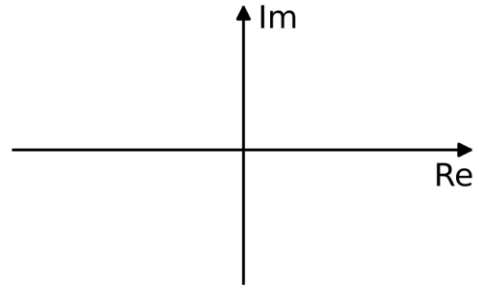


Fall b)



Pol-Nullstellen:





Sprungantwort:

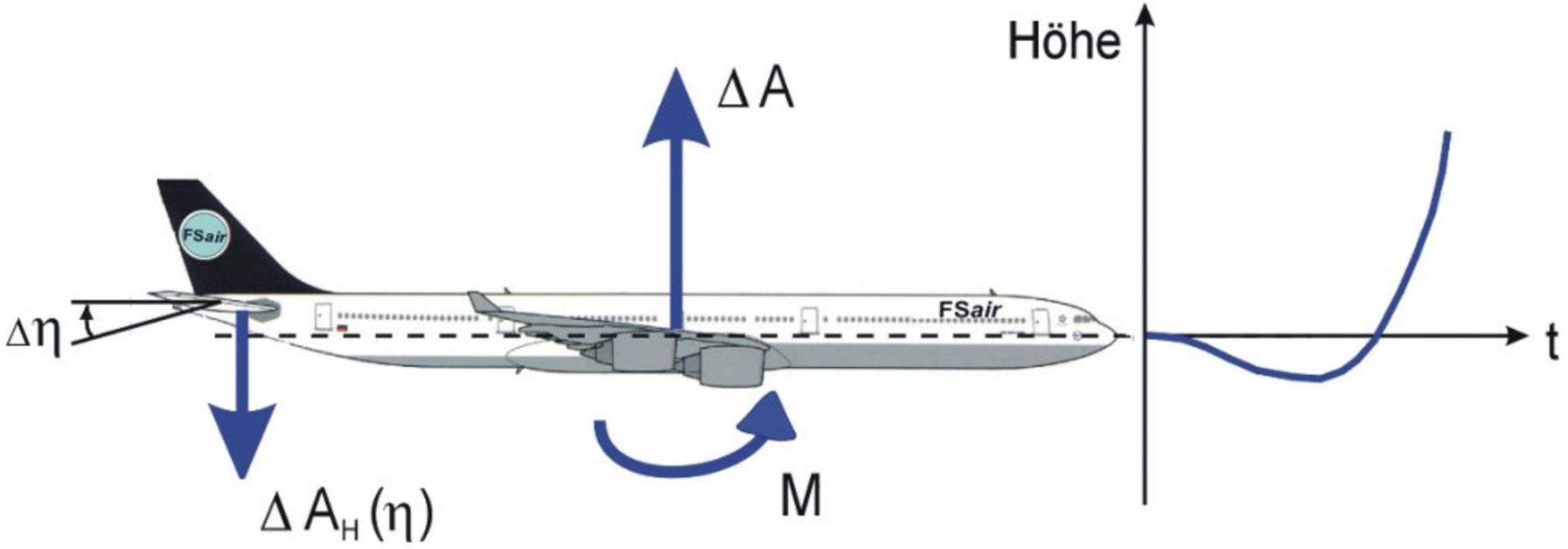




3. Pole/Nullstellen ↔ Bode

Beispiel für Allpass-Verhalten

Beispiel für Allpass – Verhalten:

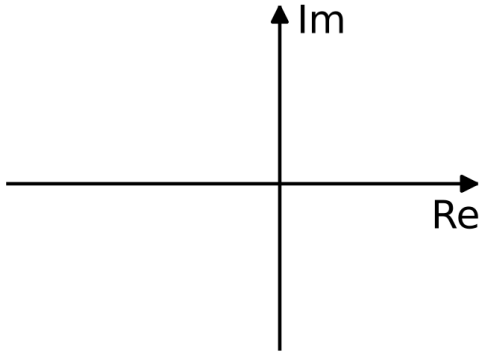


3. Pole/Nullstellen ↔ Bode

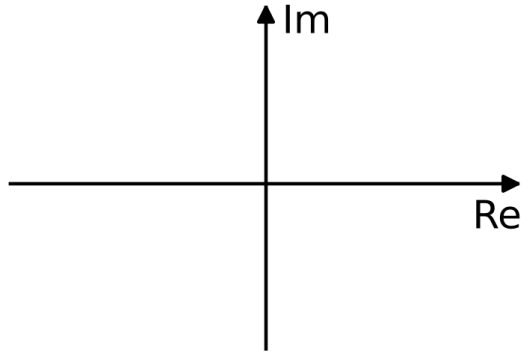
Sonderfall PDT_1

Sonderfall PDT_1:

Fall a)



Fall b)

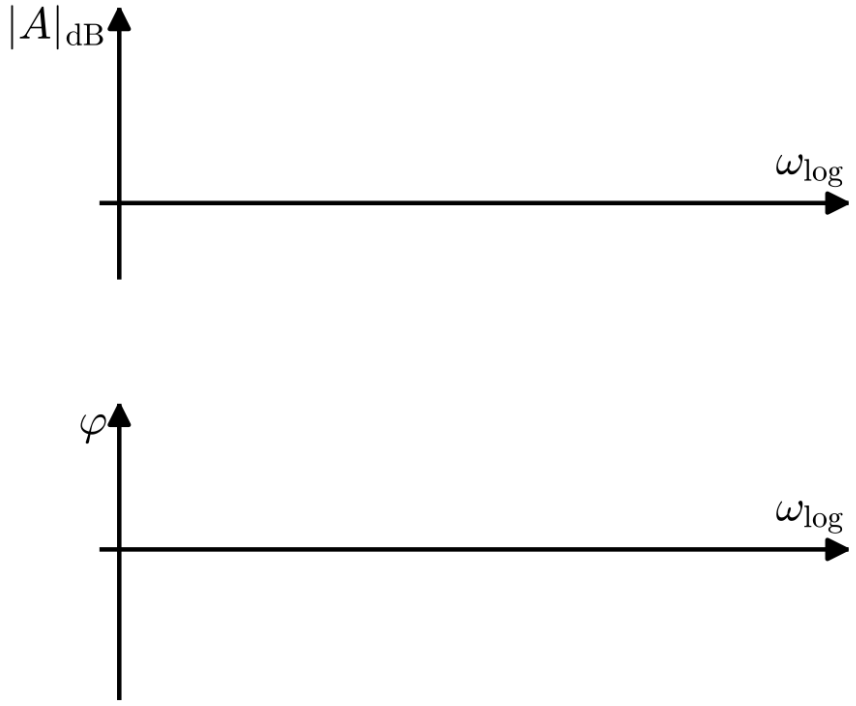


3. Pole/Nullstellen ↔ Bode

Sonderfall PDT_1: Bode-Diagramme

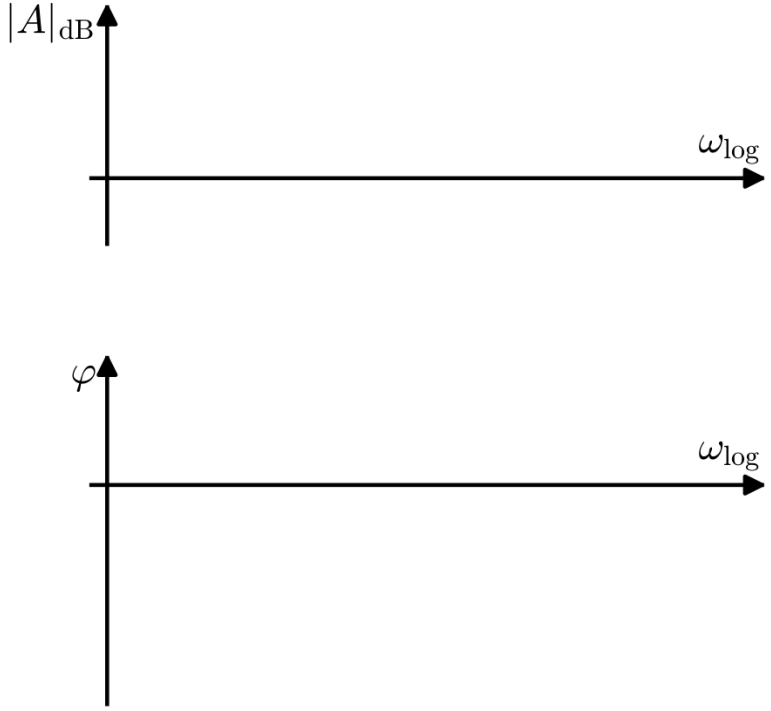
Fall a)

$$F(s) = \frac{1 + Ts}{1 + Ts} = 1$$



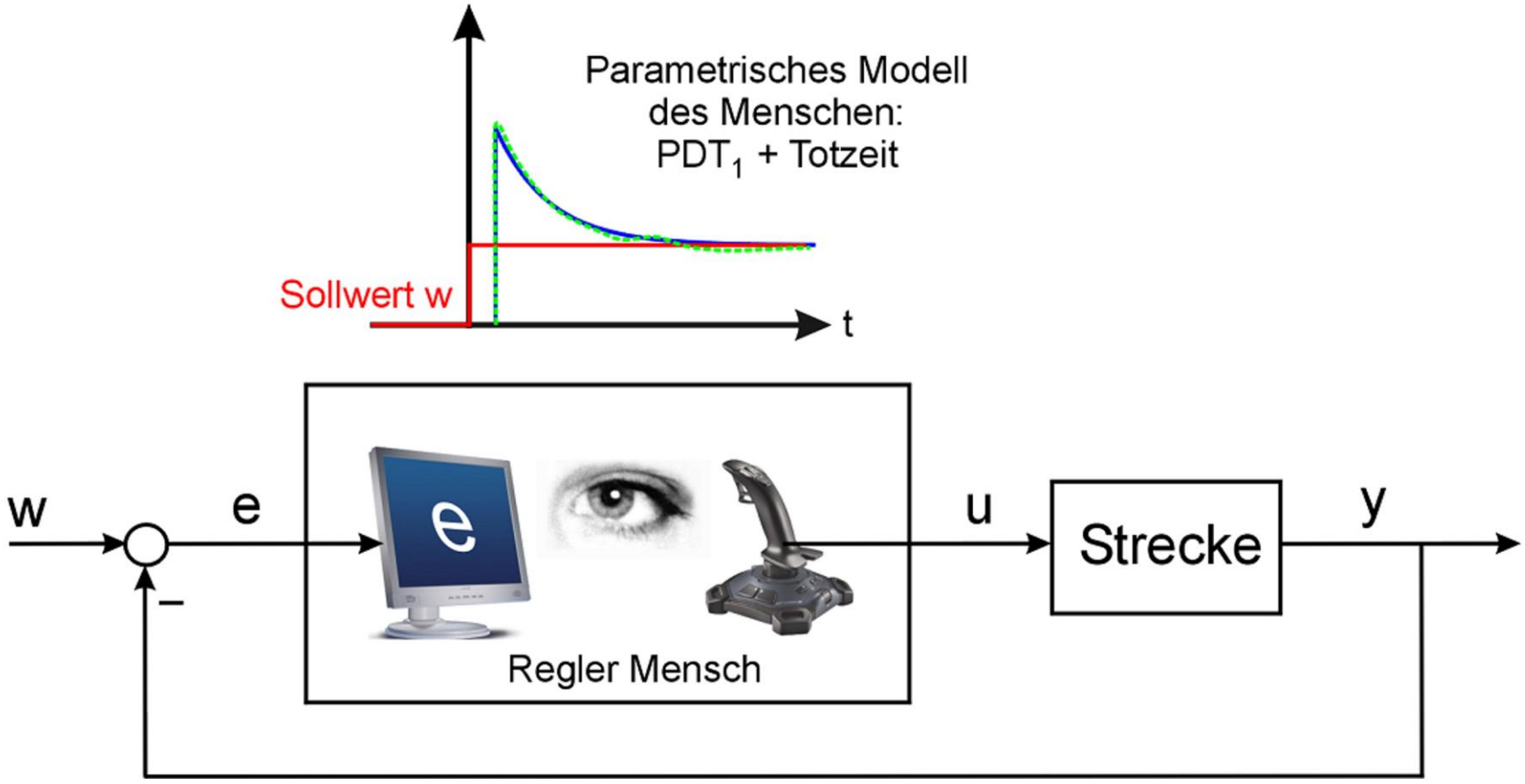
Fall b)

$$F(s) = \frac{1 - Ts}{1 + Ts}$$



3. Pole/Nullstellen ↔ Bode

Beispiel: Mensch als Regler



3. Pole/Nullstellen ↔ Bode

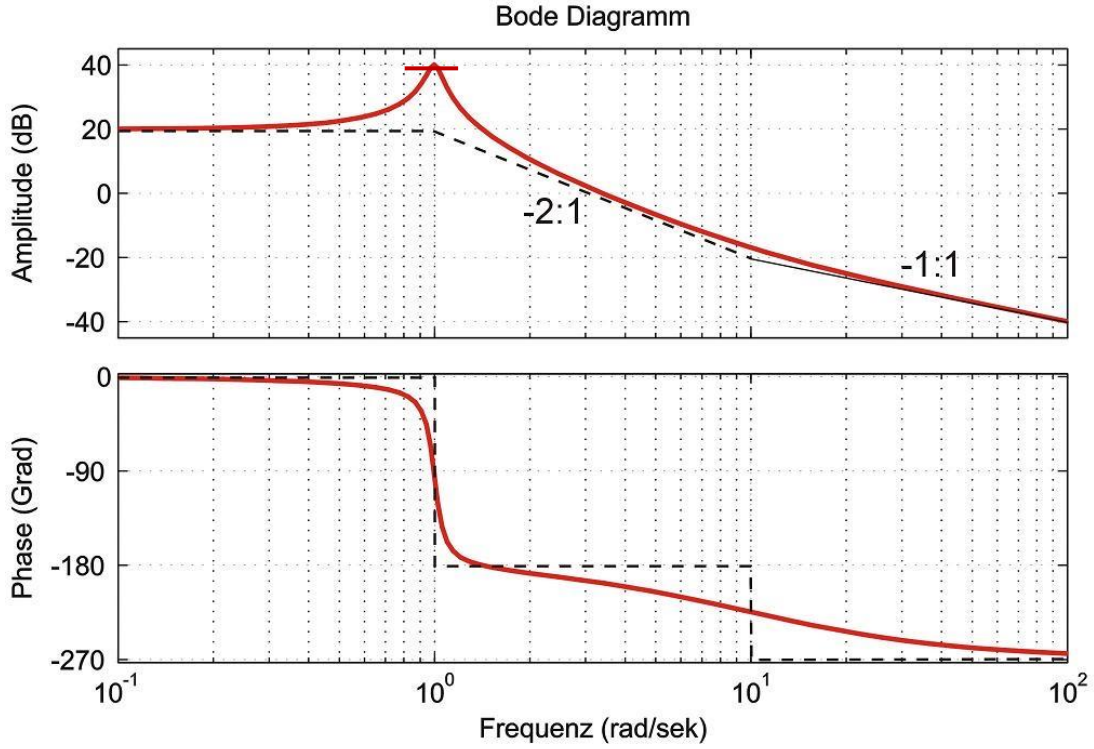
Pole vs. Nullstellen im Bode-Diagramm

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	einfacher Pol		einfache Nullstelle	
	stabil	**instabil**	**phasenminimal**	**nicht phasenminimal**
	$\mathrm{Re} < 0$	$\mathrm{Re} > 0$	$\mathrm{Re} < 0$	$\mathrm{Re} > 0$
Amplitudengang	$-1:1$ (bzw. -20 dB/Dek.)		$+1:1$ (bzw. $+20$ dB/Dek.)	
Phasengang	$-90°$	$+90°$	$+90°$	$-90°$

3. Pole/Nullstellen ↔ Bode

1. Beispiel: Übertragungsfunktion aus Bode-Diagramm (z. B. Messung)



3. Pole/Nullstellen ↔ Bode

1. Beispiel: Bestimmung der Parameter

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Übertragungsfunktion: $F(s) = K \cdot \frac{1-\frac{1}{\omega_1}s}{\frac{1}{{\omega_0}^2}s^2+\frac{2D}{\omega_0}s+1}$

Verstärkung K für $\omega \to 0$:

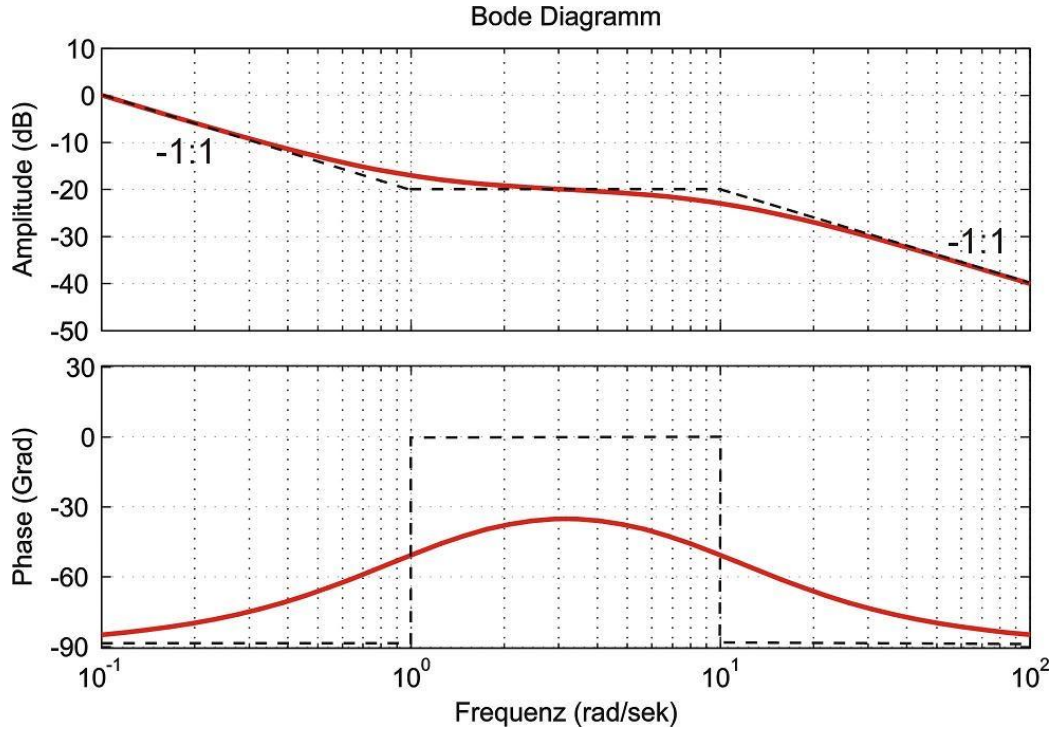
Dämpfung D:

Eckfrequenzen ω_0 & ω_1:

Einsetzen in ÜF:

3. Pole/Nullstellen ↔ Bode

2. Beispiel



Summary of Lecture

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- Non-minimum phase systems
  - Dead-time to describe time delay
  - In order to get a rational function the Padé-approximation is used
- System classification
  - Stability determined by location of poles in complex plane: System can be asymptotically stable ($Re\{\lambda_i\} < 0$), marginally stable ($Re\{\lambda_i\} = 0$) or unstable ($Re\{\lambda_i\} > 0$)
  - Systems are said to be minimum phase if zeros are in left half plane ($Re\{n_i\} \leq 0$) and non-minimum phase if zeros are in right half plane ($Re\{n_i\} > 0$)

FSR